A Major Project Report
(Phase-I)

On

NEUROSENSE AI : BRAIN HEALTH MONITORING SYSTEM

Submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IILM UNIVERSITY

Session 2026-27

in

Computer Science and Engineering

By

Shruti Jha (2410030303)

Nitesh Raj Singh (2410030254)

Nitya (2410030033)

Prashant Shekhar Pathak (2410030315)

Under the guidance of Dr. Gunjan Mittal Roy

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
IILM UNIVERSITY, GREATER NOIDA

May, 2026

SCHOOL OF COMPUTER SCIENCE AND ENGG.

IILM UNIVERSITY, GREATER NOIDA

CANDIDATE'S DECLARATION

We hereby certify that the work presented in the **Phase-I report** of the project entitled "**NEUROSENSE AI BRAIN HEALTH MONITORING SYSTEM**", in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in the **School of Computing Science and Engineering, IILM University**, Greater Noida, is an original work carried out by us under the supervision of **Dr. Gunjan Mittal Roy**, School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-I report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

Shruti Jha (241003030)
Nitesh Raj Singh (2410030254)
Nitya (2410030033)
Prashant Shekhar Pathak (2410030315)

This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Signature of Guide Name and Details of Guide

Name: Dr. Gunjan Mittal Roy

Designation: Assistant Prof.

School: School of Computing Science and Engineering

IILM University, Greater Noida

Table of Contents

SL. No	Section	Page no.
	Abstract	1
1	Problem Statement	2
2	Literature Survey	3
3	Objectives	4
4	Scope	5
5	Feasibility Study	6
	References	
	Plagiarism Report	

ABSTRACT

NeuroSense AI is a proposed AI-based brain health monitoring system designed to support the early risk screening of **Alzheimer's disease and Parkinson's disease**. Both conditions can show changes in the brain, nervous system, movement, and speech before they are formally diagnosed. However, many existing approaches mainly focus on one type of information at a time. NeuroSense AI aims to bring these different sources of information together to provide a more complete view of an individual's brain health.

The system uses **MRI and EEG** for the Alzheimer's module to study structural and neural patterns, while the Parkinson's module focuses on **speech, gait, tremor, and movement-related features**. Instead of treating all the data as one common problem, the proposed system uses separate disease-specific models and then combines their results through a common monitoring and reporting layer.

To make the results easier to understand, NeuroSense AI will include **Explainable AI techniques such as SHAP and Grad-CAM**. It will also support longitudinal monitoring, allowing changes in an individual's results to be observed over time rather than depending on a single assessment. The project will use publicly available, de-identified datasets and open-source technologies for development and evaluation.

Overall, NeuroSense AI aims to provide a practical and interpretable platform that can assist in **early risk screening and continuous brain-health monitoring**. The system is intended to support screening rather than replace professional medical diagnosis, while its modular design allows additional data sources and neurological conditions to be included in the future.

PROBLEM STATEMENT

Alzheimer's disease and Parkinson's disease have long pre-symptomatic periods during which structural, neural, motor and vocal changes may accumulate before conventional diagnosis. Existing screening approaches are commonly unimodal, relying on MRI, EEG, speech or gait independently. Their reported performance may also decrease when moving from controlled disease-versus- healthy classification to early-stage screening, differential assessment or subject-independent evaluation. Therefore, there is a need for an early-risk screening platform that combines complementary non-invasive biomarker streams, maintains disease-specific modelling, provides interpretable outputs and supports longitudinal monitoring.

Problem Statement: To design and develop NeuroSense AI, a unified, multimodal and explainable brainhealth monitoring platform that integrates structural MRI, EEG, speech and gait/movement analysis through disease-specific AI modules for Alzheimer's disease and Parkinson's disease, supported by a common longitudinal monitoring and explainability layer.

LITERATURE SURVEY

The literature reviewed for NeuroSense AI suggests that using **multiple types of neurological data together** can give a broader understanding of brain-related disorders than depending on only one source. In Alzheimer's research, MRI-based methods have been used to study structural changes in the brain, while EEG provides information about brain activity. Studies combining these approaches also show the potential of Explainable AI for making model results easier to understand. However, combining different data types can be technically challenging and may require more computational resources.

For Parkinson's disease, researchers have explored **speech, gait, and movement patterns** as possible indicators of the condition. Changes in voice characteristics can provide useful information, while gait analysis can capture differences in walking speed, stride patterns, balance, and movement. Smartphone and wearable-based approaches may make such analysis more accessible, but results can become less reliable when data are collected in uncontrolled real-world conditions.

The reviewed studies also reveal some important limitations in current research. A large portion of the work focuses on a **single disease or a single type of data**, and fewer studies consider changes in an individual over a longer period. Issues such as subject-dependent evaluation, limited explainability, and difficulties in applying complex models in real-world settings are also reported.

Based on these findings, NeuroSense AI proposes a **multimodal and disease-specific approach** that combines MRI and EEG for Alzheimer's disease and speech, gait, tremor, and movement information for Parkinson's disease. The system also aims to include Explainable AI and longitudinal monitoring so that its results are easier to interpret and changes can be observed over time.

OBJECTIVES

The major objectives of NeuroSense AI are:

1. Develop a unified multimodal AI-assisted brain-health monitoring platform.

2. Develop an Alzheimer's-specific module using MRI and EEG.

3. Develop a Parkinson's-specific module using gait, tremor, movement and speech.

4. Extract relevant structural, neural, acoustic and movement biomarkers.
3NeuroSense AI Major Project– Phase I

5. Develop and compare machine-learning and deep-learning models. 6. Implement a multimodal fusion mechanism.

7. Integrate Explainable AI techniques such as SHAP and Grad-CAM.

8. Develop longitudinal monitoring based on an individual's historical baseline.

9. Evaluate models using subject-independent validation protocols.

10. Develop an integrated brain-health dashboard/report.

NeuroSense AI aims to develop a **multimodal brain-health monitoring system** that combines different neurological data sources for early risk screening. It will use **MRI and EEG for Alzheimer's disease** and **speech, gait, tremor, and movement data for Parkinson's disease**. The project will compare AI models, combine information from multiple modalities, and use **Explainable AI** to make results easier to understand.

The system will also track changes over time using an individual's previous data, evaluate models using reliable validation methods, and present the results through an **integrated brain-health dashboard/report**. Overall, the goal is to create a practical and understandable platform for monitoring brain-health risks.

SCOPE

NeuroSense AI focuses on developing a **multimodal brain-health monitoring platform** for the early risk screening of Alzheimer's and Parkinson's disease. The system will use different types of information to provide a broader assessment instead of depending on a single data source.

The scope includes **MRI and EEG analysis for Alzheimer's**, along with **speech, gait, tremor, and movement analysis for Parkinson's**. It will also cover feature extraction, multimodal data fusion, Explainable AI, longitudinal monitoring, and the generation of a Brain Health Indicator. The processed results will be presented through a dashboard for easier visualization and reporting.

The project is designed in a modular way, so additional features can be added later without rebuilding the entire system. Future possibilities include **handwriting analysis, facial-expression analysis, wearable signals, cognitive assessment, other neurodegenerative disorders, and mobile or edge-based deployment**.

Overall, the scope is to create an interpretable and expandable AI-based platform that can support brain-health screening and monitoring, while keeping its results as screening indicators rather than medical diagnoses.

FEASABILITY STUDY

1 .Technical Feasibility

The project is technically achievable because suitable public datasets and open-source tools are available for the different data types. Technologies such as **Python, PyTorch/TensorFlow, MONAI, MNE-Python, and librosa** can support the development and processing of the proposed system. The modular structure also makes it possible to build and test the Alzheimer's and Parkinson's modules separately before combining them.

2. Data Feasibility

The project can make use of **publicly available and de-identified research datasets** for academic development. However, the datasets may differ in terms of participant characteristics, data collection methods, sampling rates, and file formats. Because of these differences, proper preprocessing and data standardization will be important before using the data for model development.

3. Economic Feasibility

The development cost is expected to be relatively low because the project can rely mainly on **public datasets and open-source software**. University computing facilities can be used for development, while free or affordable cloud GPU services can be considered when additional computing power is needed.

4. Operational Feasibility

The system can be developed step by step, with the Alzheimer's and Parkinson's modules being worked on independently. This approach makes the development process easier to manage, allows individual components to be tested properly, and reduces the difficulty of integrating the complete system later.

5. Ethical Feasibility

Since the project deals with health-related information, **only de-identified research data** should be used during development. The system must also clearly state that its output is intended for **risk screening and monitoring only and should not be considered a medical diagnosis**.

REFERENCES

1. Research literature on multimodal artificial intelligence for Alzheimer's disease using MRI, EEG and clinical information.
(https://www.sciencedirect.com/science/article/pii/S0031320326014366?utm_source=chatgpt.com)

2. Research literature on deep-learning-based Alzheimer's disease detection using structural MRI.
(https://www.sciencedirect.com/science/article/pii/S3050577126000253?utm_source=chatgpt.com)

3. Research literature on explainable multimodal AI for Alzheimer's disease.
(https://www.sciencedirect.com/science/article/pii/S0031320326014366?utm_source=chatgpt.com)

4. Research literature on explainable EEG-based machine learning for Alzheimer's disease and cognitive assessment.
(https://pubmed.ncbi.nlm.nih.gov/42297902/?utm_source=chatgpt.com)
(https://www.nature.com/articles/s41598-026-57069-1?utm_source=chatgpt.com)

5. Research literature on EEG spectral features and deep learning for Alzheimer's disease.
(https://pmc.ncbi.nlm.nih.gov/articles/PMC13315799/?utm_source=chatgpt.com)

6. Research literature on smartphone-based Parkinson's disease detection using voice, finger tapping and gait.
(https://pubmed.ncbi.nlm.nih.gov/40325040/?utm_source=chatgpt.com)

7. Comprehensive review literature on artificial intelligence for Parkinson's disease.
(https://www.nature.com/articles/s41531-025-01025-9?utm_source=chatgpt.com)

8. Research literature on graph convolutional networks for Parkinson's gait analysis.
(https://pubmed.ncbi.nlm.nih.gov/31351213/?utm_source=chatgpt.com)

9. Research literature on machine-learning-based Parkinson's detection and severity classification using gait.
(https://www.nature.com/articles/s41598-024-83975-3?utm_source=chatgpt.com)

10. Research literature on wearable sensor-based quantitative gait analysis in Parkinson's disease.
(https://pubmed.ncbi.nlm.nih.gov/29020012/?utm_source=chatgpt.com)
(https://pmc.ncbi.nlm.nih.gov/articles/PMC5636070/?utm_source=chatgpt.com)

11. Research literature on objective gait and posture assessment using wearable sensors.
(https://pubmed.ncbi.nlm.nih.gov/40860216/?utm_source=chatgpt.com)

12. PADS smartwatch and smartphone dataset research for Parkinson's disease analysis.
(https://www.physionet.org/content/parkinsons-disease-smartwatch/1.0.0/?utm_source=chatgpt.com)

13. Systematic review literature on wearable sensors for neurodegenerative disease monitoring.
(https://pmc.ncbi.nlm.nih.gov/articles/PMC10192816/?utm_source=chatgpt.com)